



T.K.M COLLEGE OF ENGINEERING

Kollam – 691005

24MEJ404-Heat Transfer & Thermal Machines

PROJECT REPORT

ANALYSIS OF HEAT TRANSFER THROUGH COMPOSITE WALLS FOR MAXIMUM THERMAL EFFICIENCY AND ENERGY CONSERVATION USING ANSYS

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ABSTRACT

The increasing demand for energy in the building sector has made thermal efficiency a critical factor in modern design, as a major portion of consumption is driven by heating and cooling loads influenced by heat transfer through building envelopes. To address this, our project focuses on analyzing heat transfer through composite wall structures to enhance thermal performance using Ansys. In this phase, we conducted a detailed steady-state thermal analysis in ANSYS, comparing conventional and composite configurations using materials like concrete blocks, cement plaster, glass wool, and a cement-water hyacinth composite. By inputting standard thermal conductivity values and applying specific boundary conditions, we used heat flux as the primary metric to evaluate performance. The simulation results show that the composite wall assemblies exhibit significantly lower heat flux than conventional ones, proving that the usage of insulating layers effectively boosts thermal resistance. Furthermore, we used **OpenStudio** and the **EnergyPlus** engine to model a **50 m² office building**, with the specific weather data of **Thiruvananthapuram**. This allowed us to calculate and compare the actual annual site energy consumption between a standard conventional wall and our composite wall designs. The simulation results confirmed that the final **4-layer composite wall** significantly outperforms conventional wall, decreasing the annual energy demand from **14,592 kWh** to **10,525 kWh**. While our 5-layer walls showed slightly higher savings (down to 9,714 kWh), we chose the 4-layer wall as our final design. As an engineering students, we had to **consider the economic aspects**, the 4-layer wall offers the best balance of high thermal performance without adding extra material costs, excessive wall thickness and construction complexity of a 5-layer wall. By achieving a substantial **27.8% reduction** in energy usage, this project proves that our wall system is a practical and sustainable solution for cutting down cooling loads in tropical climates like Kerala.

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INTRODUCTION

Today, the building sector is one of the biggest drivers of global energy consumption, which has a massive impact on climate change and the rise of greenhouse gas emissions. Here in India, buildings account for roughly 30% to 40% of our total energy usage, and that demand is growing by about 8% to 9% every single year. Most of this power goes toward HVAC systems just to keep indoor spaces comfortable against the outside heat. We found that while making air conditioners more efficient is helpful, the real long-term solution is to design better building envelopes or walls that stop the heat from getting indoor.

The primary objective of this project was to model and analyze the performance of a multi-layer composite wall designed to reduce heat transfer and minimize overall energy consumption. Traditional wall systems, which consist of a single layer of concrete blocks ($k = 1.5 \text{ W/mK}$) finished with cement plaster on both sides ($k = 0.75 \text{ W/mK}$), offer good structural strength but suffer from high thermal conductivity. This leads to excessive heat gain during summer and heat loss during winter, forcing HVAC systems to work harder. To solve this engineering challenge, we designed a composite wall by using high-resistance insulating materials that act as thermal barriers. Specifically, we utilized **Glass-Wool** ($k = 0.045 \text{ W/mK}$) and a sustainable, bio-based **Cement-Water Hyacinth composite** ($k = 0.0765 \text{ W/mK}$), which not only provides insulation but also promotes the use of eco-friendly building materials.

Our methodology involved a detailed computational approach to evaluate the thermal performance of our proposed designs. We first used **ANSYS** to conduct steady-state thermal analysis to see how the heat would move through the wall and to measure the flow of heat through the materials. This allowed us to visualize the temperature gradient across the wall and measure the heat flux passing through different material interfaces, verifying that our insulating layers were effectively resisting heat movement. We could also see how different thicknesses and layers of materials would work before we built anything.

For the final evaluation of our project, we moved beyond basic thermal analysis to simulate real-world energy impact. We used OpenStudio to simulate the energy use of a small office building of 50m² area. We compared our composite wall design to a conventional wall and found that our design used a less energy. Our final design had four layers: Cement-Water Hyacinth, Glass-Wool, Concrete and Plaster. Our final simulation results showed that this 4-layer configuration successfully reduced the annual site energy consumption from **14,592 kWh** to **10,525 kWh** which is 27.8 percent less energy than the standard wall. This shows that using the materials and design can make a big difference in reducing energy use and making buildings more sustainable.

OBJECTIVES

The primary objective of this project is to analyze and improve the thermal performance of building walls by designing and evaluating composite wall structures using numerical simulation techniques.

The specific objectives of the project are as follows:

1. Selection and characteristic study of each material for the composite wall composition.
2. Modelling and thermal analysis of the multi-layer composite wall using Ansys Workbench.
3. Development of the composite wall and investigating the effect of varying materials and layer thickness of each materials.
4. Design and energy simulation of the composite wall for building applications using OpenStudio and EnergyPlus

By completing these objectives, we established a clear understanding of heat transfer in composite systems and proved that our **4-layer wall** is a practical, high-performance solution for modern construction.

SCOPE

Scope of the Project

The scope of this project focuses on the thermal optimization of building wall to address the rising energy demands in the construction sector. While there are many ways to save energy in a building, our study specifically targets the heat transfer through walls, as they represent the largest surface area exposed to the environment.

- **Material Research and Hybrid Design:** We limited our study to four specific materials: Concrete, Cement Plaster, Glass-Wool, and a bio-based Cement-Water Hyacinth composite. The scope includes analyzing how these materials interact when layered together to form a thermal barrier.
- **Computational Thermal Analysis:** The project utilizes **ANSYS** to simulate steady-state heat transfer. This is limited to internal temperature distribution and heat flux calculations to validate the effectiveness of the composite layers before full-scale application.
- **Whole-Building Energy Modelling:** Using **OpenStudio** and the **EnergyPlus** engine, we expanded our scope from a single wall section to a complete 50 m² office building model. This allowed us to predict annual energy consumption and compare the efficiency of our 4-layer design against a conventional wall.
- **Sustainability and Environmental Impact:** We are using Water Hyacinth this helps with sustainability and environmental impact of engineering. We want to make sure that our buildings are sustainable for the environment. We are tried to reduce the carbon footprint of buildings by using Water Hyacinth-Cement composite as a insulating material.

LITERATURE REVIEW

SI NO	Name of author	Name of Journal / year	Title	Work	Finding
1	Jing Sun, Qiuqi Zhang, An Chen, and Zhili Wu.	Journal of Building Engineering Volume 91, 15 August 2024, 109614	Energy performance of buildings with multi-ribbed composite wall structural system.	- Evaluate the energy saving effect of multifunctional multi-ribbed composite wall (MRCW) structural system. - They compare standard wall with composite wall(MCRW) using Energyplus software.	- The total energy consumption of the MRCW system can be reduced by 11.4 % compared with that of the shear wall system. - MRCW system cannot only bear load, but also saves energy
2	Biswajit Nath, Sujit Roy, Biplab Das, and M.A. Ehyaei	Thermal Science and Engineering Progress Volume 53, August 2024, 102751	Experimental and numerical study of a novel composite building wall U-value.	- The researchers tested different types of bricks to find the best one for blocking heat. - Then built a composite wall by combining the best brick with glass wool insulation and an air gap.	- The indoor temperature is reduced by 40.83% with composite wall. - The maximum heat flux through composite wall is reduced by 3.37%. - he average U-value obtained using composite wall is 0.51 W/m² K.(a measure of how much heat leaks through a wall).

SI NO	Name of author	Name of Journal / year	Title	Work	Finding
3	Yanli Hu	Chemical Engineering Transactions Vol. 51, 2016	Analysis on Thermal Performance of Composite Wall of Energy-saving Buildings	3D temperature field and thermal resistance of new energy-saving wall blocks are calculated with finite difference method in order to save land resources and reduce the engineering cost.	It is shown in the research that new composite walls have good effects on the winter and the summer, and its thermal stability and indoor comfort are increased significantly compared with traditional <u>non energy-saving</u> walls.
4	Yushun Li, Jian Yao, Ran Li, Zhenwen Zhang, and Jialiang Zhang.	Energy and Buildings Volume 156, 1 December 2017, Pages 225-237	Thermal and energy performance of a steel-bamboo composite wall structure	Thermal and energy performance of a novel steel-bamboo wall was investigated and U value, time lag and decrement factor of this wall was determined and the result was compared with traditional wall structure.	- An improvement of U value by up to 26.1%–48.4%. - The steel-bamboo wall reduced total energy use by 27% to 41% compared to traditional walls.

METHODOLOGY

In this composite wall heat transfer project, the methodology focuses on modelling and analysing different wall configurations using ANSYS to understand the thermal behaviour of each wall configuration. The goal at this stage is to establish a foundational understanding of how composite walls can improve thermal efficiency compared to conventional walls.

Material Selection and Initial Characterization: We start by selecting a range of materials with differing thermal conductivities, including concrete block, cement plaster, cement–water hyacinth composite, and glass wool. Each material is chosen for its potential to contribute to improved insulation and also structural strength when used in composite walls.

Modeling Composite Wall Configurations: Using ANSYS, we construct digital models of both conventional and composite walls. The composite walls are modeled by layering the selected materials in various configurations to assess how different combinations impact heat transfer.

Conducting Steady-State Thermal Analysis: We perform steady-state thermal simulations to observe and measure heat flux through the different wall configurations. The focus is on capturing interim data to understand which combinations of materials provide the best thermal resistance. This analysis serves as a preliminary step to identify configurations for further refinement

Energy Consumption Analysis: This moved the project from a single wall section to a full building application.

- **Localized Weather Data:** We localized the simulation to **Thiruvananthapuram, Kerala**. We imported the **EPW (EnergyPlus Weather) file** for Thiruvananthapuram to account for the local humidity, solar radiation, and temperature cycles.
- **Building Modelling:** Using the **OpenStudio** interface and the **EnergyPlus** engine, we used a **50 m² office building**.
- **Zone Definition:** We defined thermal zones and set the internal gains (heat from people, lights, and equipment).

- **Simulation Run:** We assigned our 4-layer composite wall to the building's exterior and ran an annual simulation (8,760 hours). The software calculated the electricity required by the HVAC system to keep the building at a range of 23.6°C to 26.4°C throughout the year.

Comparative Data Analysis: In the final stage, we compared the data from the **Conventional Wall** simulation against our **4-Layer Composite Wall** simulation.

- We extracted the **Total Site Energy** consumption in kilowatt-hours (kWh).
- The conventional wall consumed **14,592 kWh** annually.
- Our final 4-layer wall configuration consumed only **10,525 kWh**.
- This allowed us to calculate the final energy savings of **27.8%**.

Selected Materials for the Composite Wall

The selection of materials plays a crucial role in determining the thermal performance of composite wall systems. In this project, materials were chosen based on their thermal conductivity, availability, structural suitability, and relevance to sustainable building practices. The objective was to combine structural materials with effective insulating layers to reduce heat transfer through the wall.

1. Concrete block was selected as the primary structural material due to its widespread use in conventional building construction. It provides adequate mechanical strength and durability; however, it has relatively high thermal conductivity, which makes it less effective as a thermal insulator when used alone.
2. Cement plaster was used as the surface finishing layer on outer sides of the wall. It improves surface uniformity, protects the wall structure, and contributes marginally to thermal resistance. Cement plaster is commonly used in buildings and was included to maintain realistic wall configurations.
3. Glass wool was chosen as an insulating material because of its low thermal conductivity and proven effectiveness in reducing heat transfer. It contains air pockets within its fibrous

structure, which significantly restrict heat flow. Glass wool is widely used in building insulation and serves as a benchmark insulating material in this study.

4. Cement–water hyacinth composite in the ratio 40:60 was selected as a sustainable and eco-friendly insulating material panel. Water hyacinth fibres possess good thermal insulation properties due to their porous structure. Incorporating this bio-based material aligns with the objective of developing energy-efficient and environmentally sustainable building solutions.

Thermal Properties of Selected Materials

Thermal conductivity is a key parameter that governs the rate of heat transfer through building materials. In this study, thermal conductivity values of the selected materials were obtained from standard literature and used as input data for steady-state thermal analysis in ANSYS. Materials with lower thermal conductivity offer higher resistance to heat flow and are therefore more effective for thermal insulation.

The thermal conductivity values of the materials used in the composite wall are as follows:

- Concrete block: 1.5 W/m·K
- Cement plaster: 0.75 W/m·K
- Glass wool: 0.045 W/m·K
- Cement–water hyacinth composite: 0.075 W/m·K

Concrete block and cement plaster exhibit relatively high thermal conductivity, resulting in greater heat transfer when used in conventional wall systems. In contrast, glass wool and cement–water hyacinth composite possess very low thermal conductivity, making them effective insulating materials.

By incorporating low-conductivity materials between structural layers, the overall thermal resistance of the composite wall is significantly improved. These thermal property values were assigned to the respective layers during simulation to accurately evaluate heat flux and thermal performance in this interim phase of the project.

OBSERVATION AND RESULT

Simulations were performed to compare a Conventional Wall against a different types of Composite Wall. The following observations were recorded during the heat flux analysis and temperature analysis using Ansys and the Energy consumption usage using Openstudio. We selected Thiruvananthapuram as the location to find the energy consumption in the Openstudio using Energyplus.

CONVENTIONAL WALL PERFORMANCE

The baseline wall, consisting of a 200mm concrete block and cement plaster, exhibited the highest Total Heat Flux. The high thermal conductivity of the concrete allowed rapid heat transfer from the external environment to the interior and heat flux obtained was 73.16 W/m^2

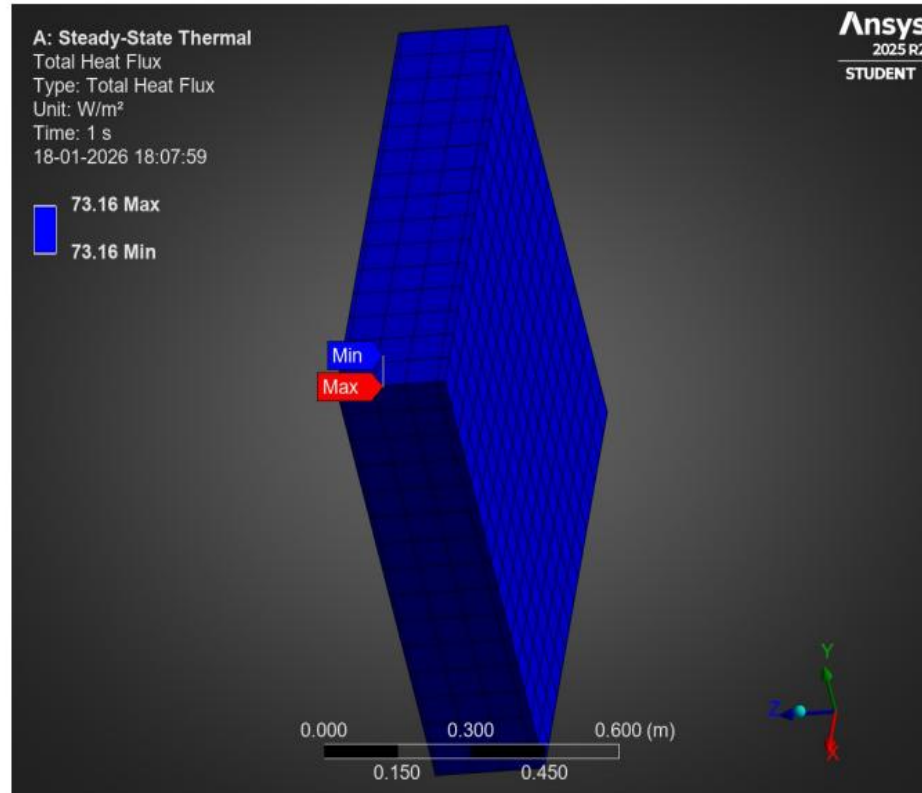


Fig. 1.1 Heat flux of Conventional Wall

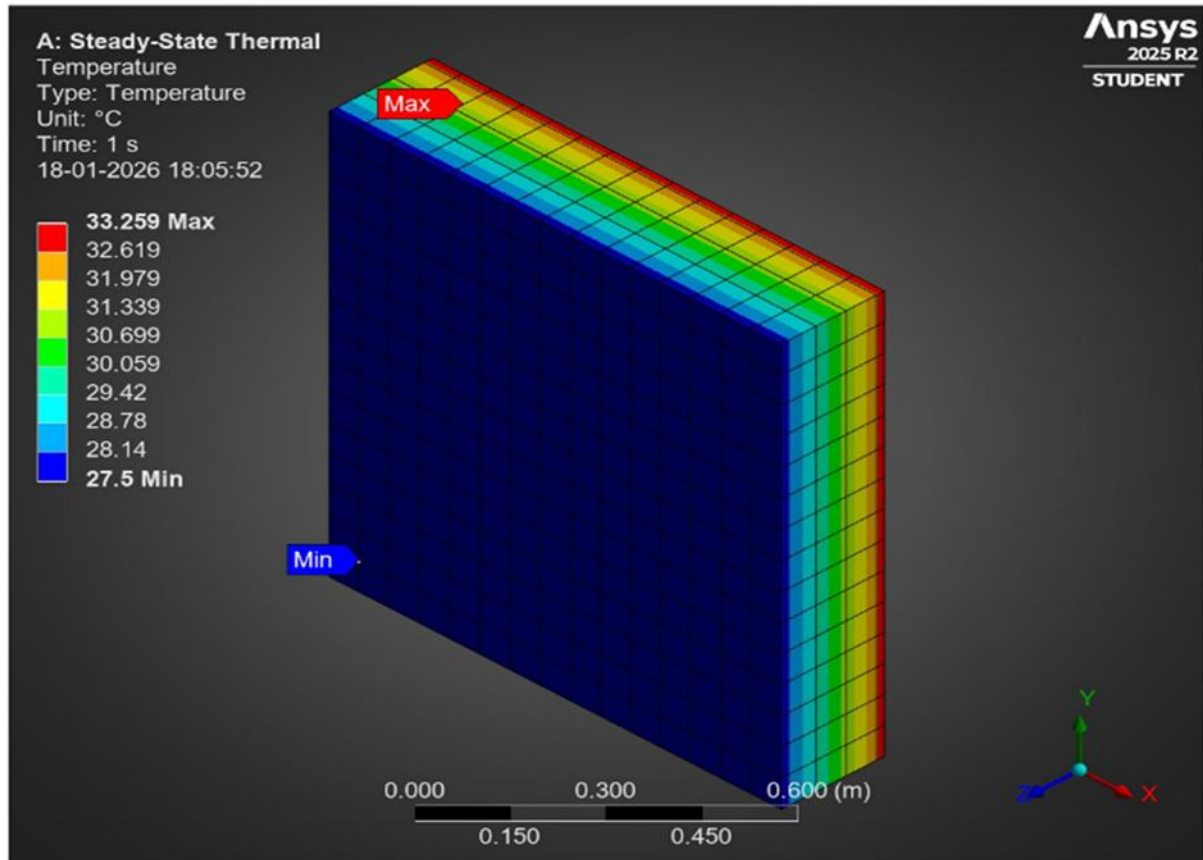


Fig. 1.2 Temperature Distribution of Conventional Wall

Data	Value
Building Name	Office Building
Total Site Energy	14,592kWh
Total Building Area	50m ²
Total Site EUI	291.83kWh/m ²

Table. 1.1 Summary Annual Site Energy Data for Office Building Model of Conventional Wall

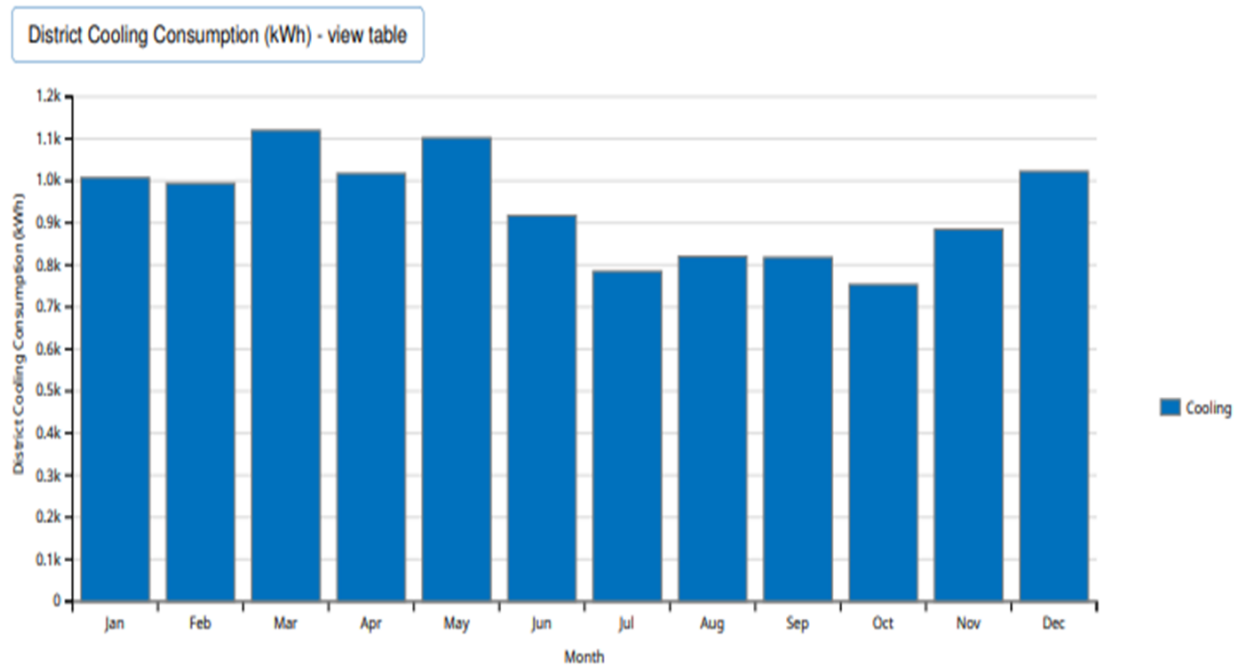


Fig. 1.3 Conventional Wall Energy Consumption throughout the year

COMPOSITE WALL PERFORMANCE

The addition of the 60:40 Water Hyacinth-Cement combination and Glass Wool layers created a significant thermal barrier.

1)4 Layers Composite Wall:

The first combination is the 4 layers wall with water hyacinth, glass wool, concrete block, cement plastering arranged from inner to outer side and heat flux obtained was 19.427W/m^2

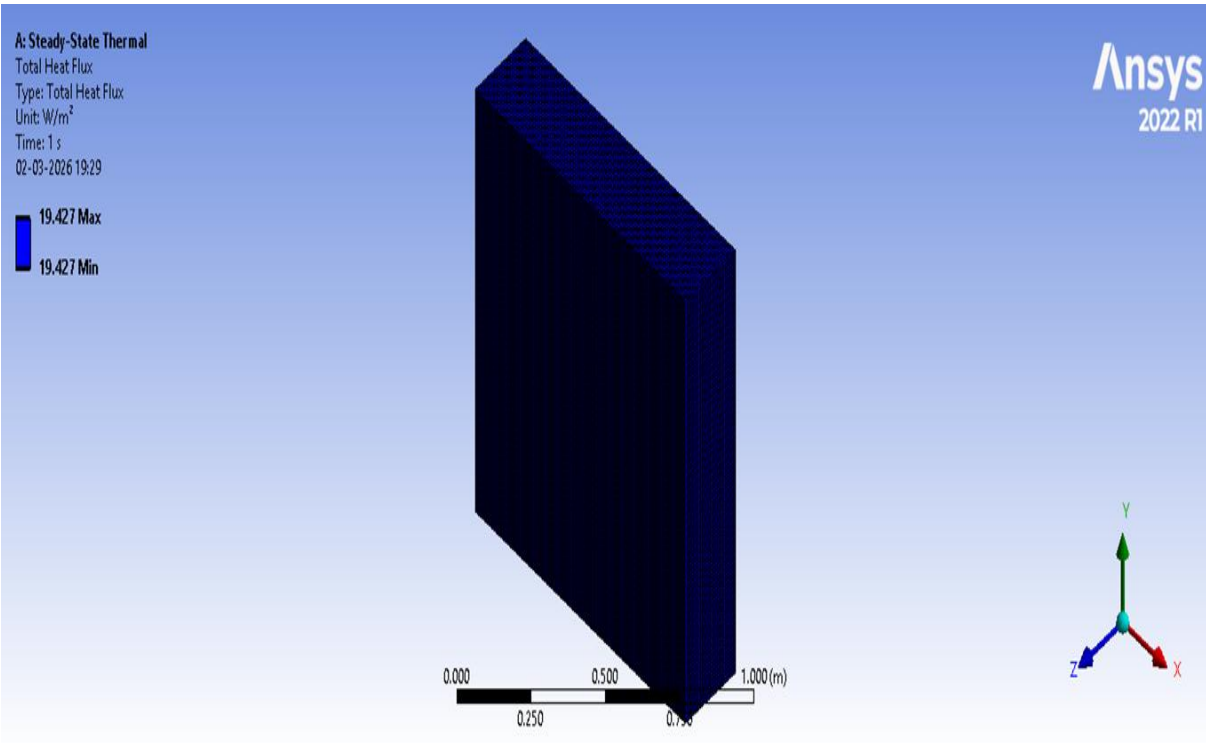


Fig. 1.4 Heat flux of 4 layers Composite Wall

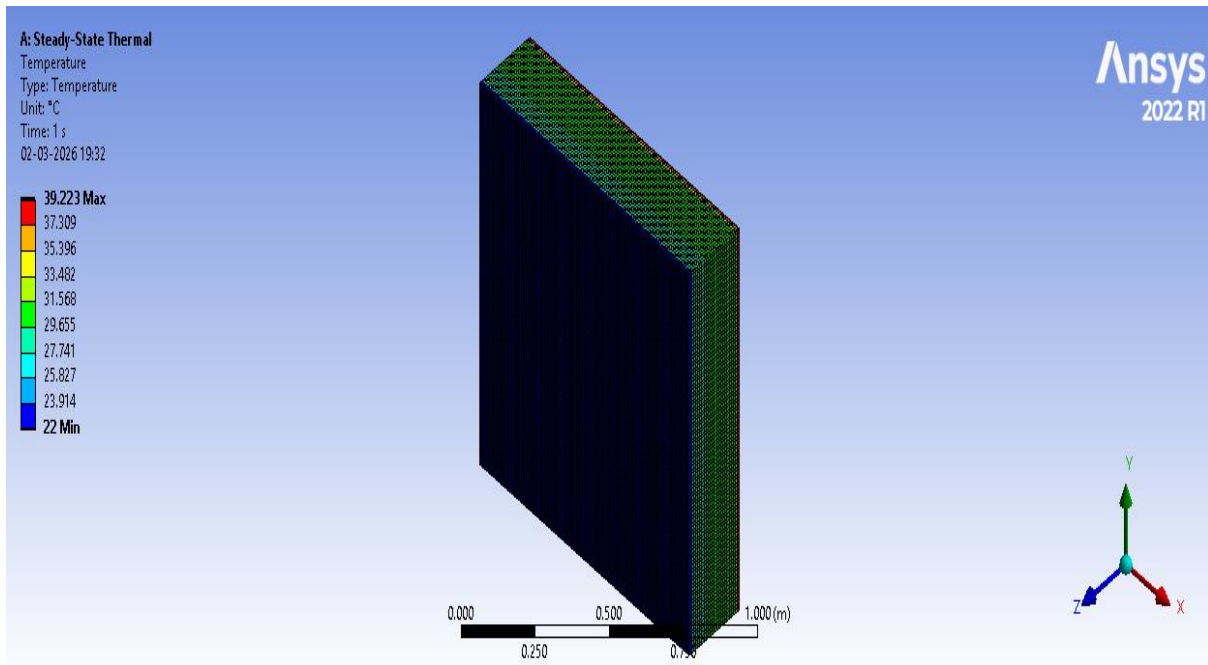


Fig. 1.5 Temperature Distribution of 4 Layer composite wall

Data	Value
Building Name	Office Building
Total Site Energy	10,525kWh
Total Building Area	50m ²
Total Site EUI	210.50kWh/m ²

Table. 1.2 Summary of Annual Site Energy Data for Office Building Model of 4 Layer composite wall

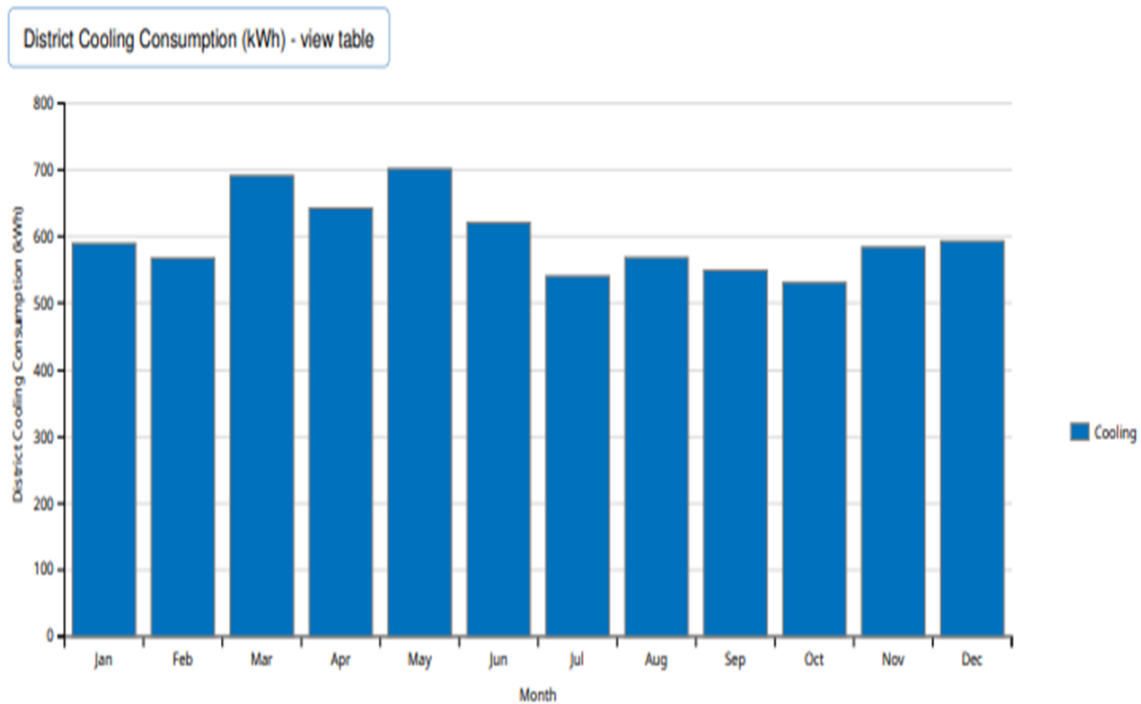


Fig. 1.6 Energy consumption of 4 Layers composite wall throughout year

2) 5 Layers Composite Wall-1

The second combination is the 5 layers wall with cement-water hyacinth, cement-water hyacinth, glass wool, concrete block, cement plastering arranged from inner to outer side and heat flux obtained was 17.023 W/m^2

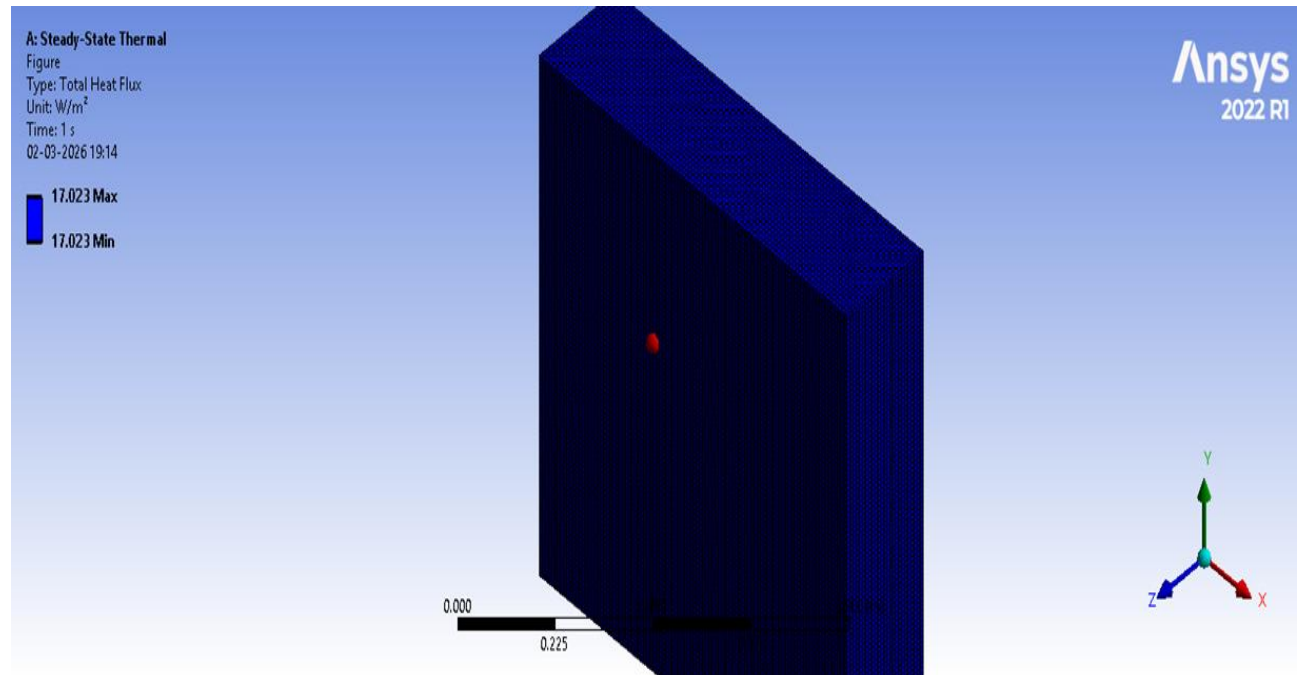


Fig. 1.7 Heat flux of 5 Layers Composite wall-1

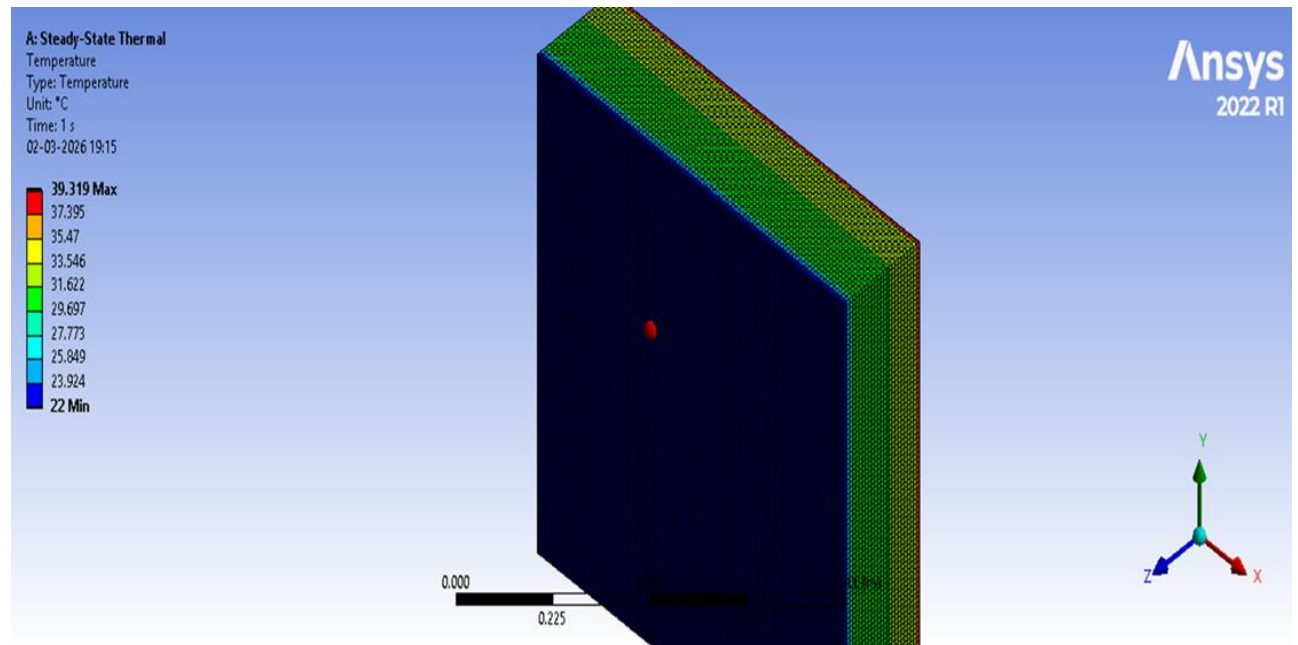


Fig. 1.8 Temperature Distribution of 5 Layer Composite Wall-1

Data	Value
Building Name	Office Building
Total Site Energy	9,986kWh
Total Building Area	50m ²
Total Site EUI	199.72kWh/m ²

Table. 1.3 Summary of Annual Site Energy Data for Office Building Model of 5 Layer Composite Wall-1

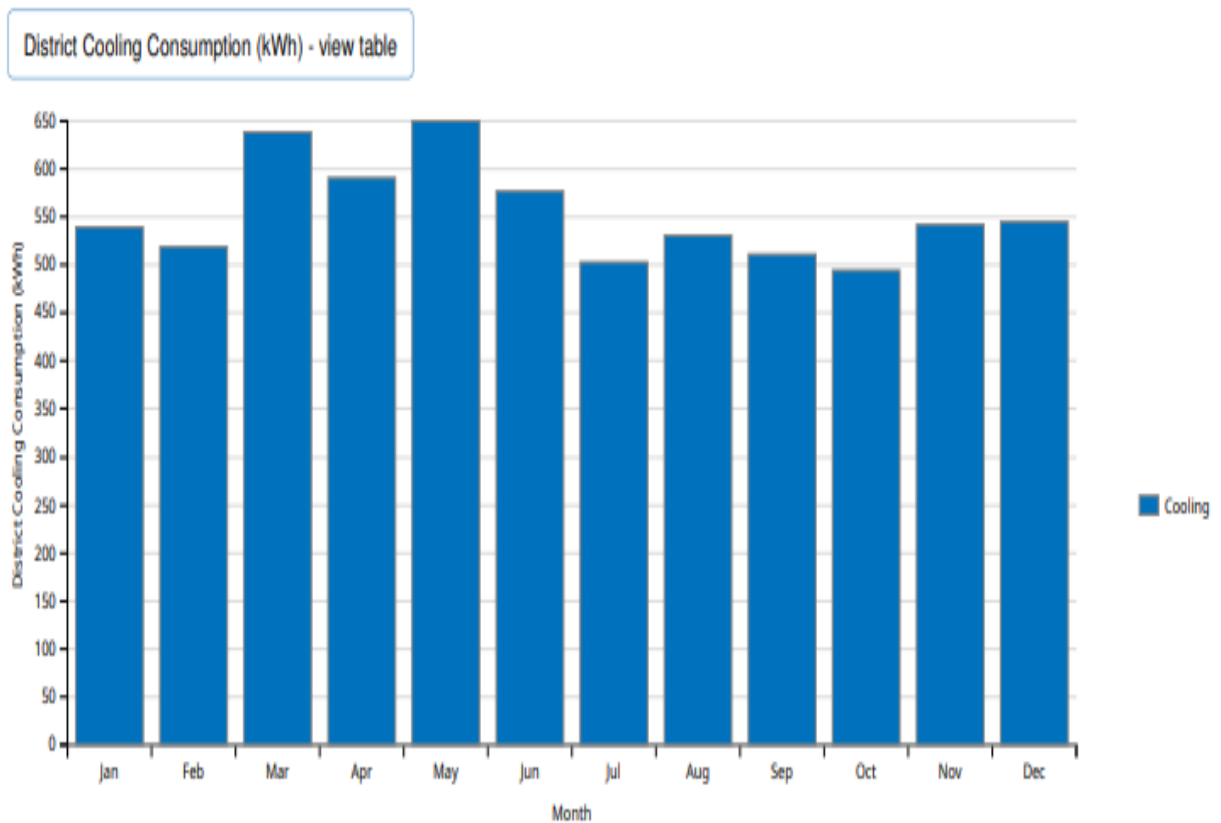


Fig. 1.9 Energy consumption of 5 Layers composite wall-1 throughout year

3) 5 Layers Composite Wall-2

The third combination is the 5 layers wall with cement-water hyacinth, glass wool, glass wool, concrete block, cement plastering arranged from inner to outer side which was found to be the lowest of the three types of walls with a heat flux of 13.399 W/m^2

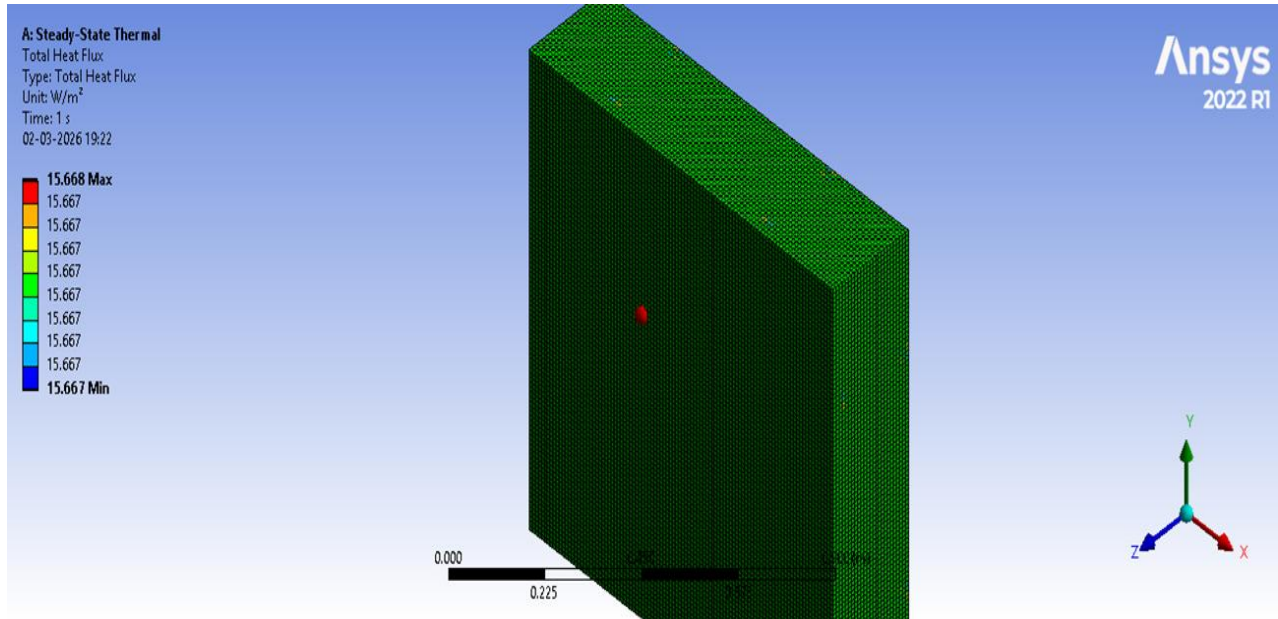


Fig. 2.0 Heat Flux of 5 Layers Composite Wall-2

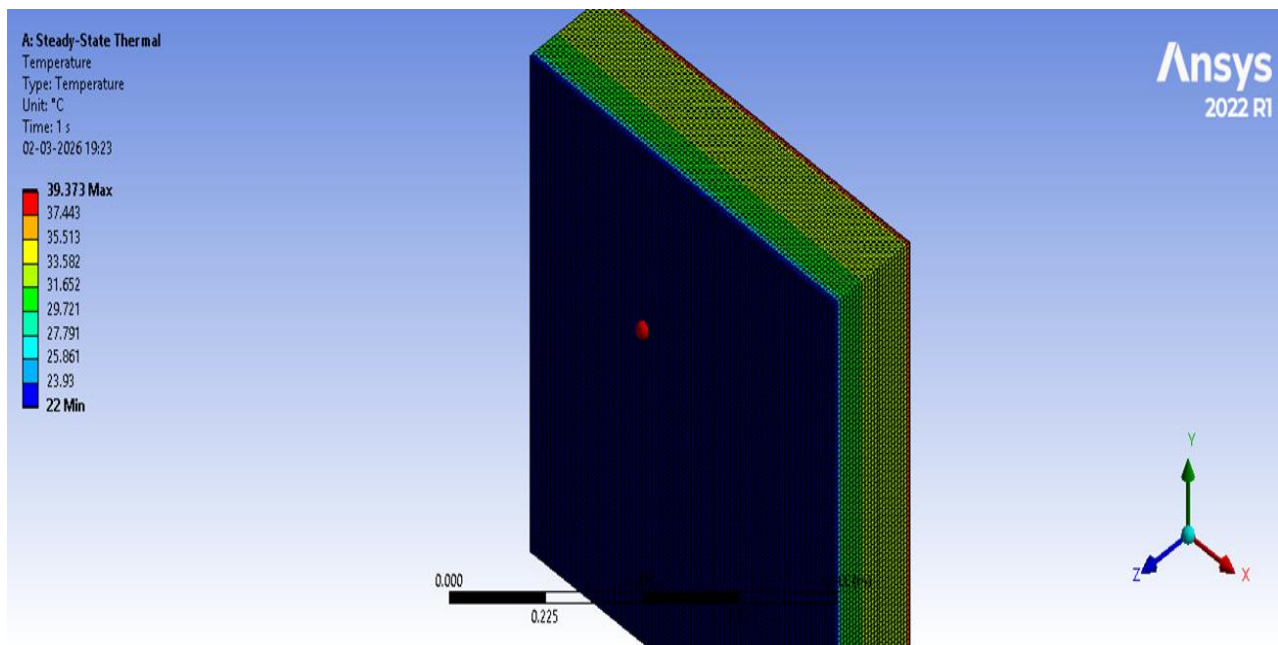


Fig. 2.1 Temperature Distribution of 5 Layers Composite Wall-2

Data	Value
Building Name	Office Building
Total Site Energy	9,714kWh
Total Building Area	50m ²
Total Site EUI	194.28kWh/m ²

Table. 1.4 Summary of Annual Site Energy Data for Office Building Model of 5 Layers Composite Wall-2

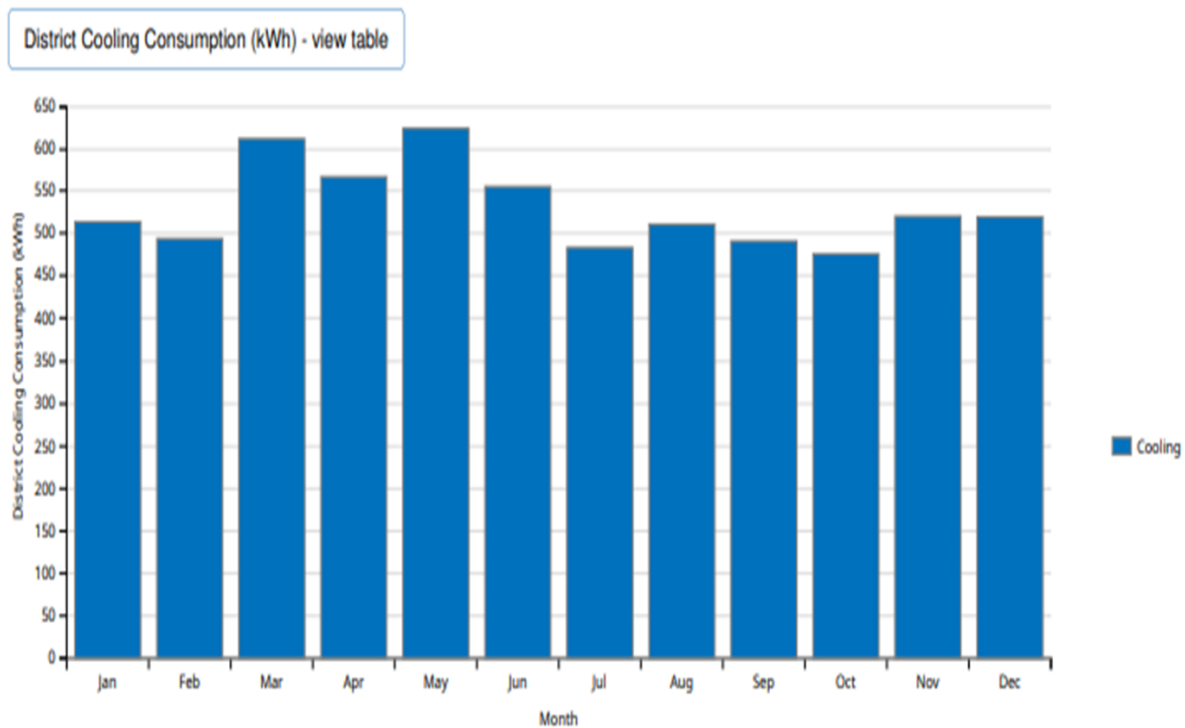


Fig. 2.2 Energy consumption of 5 Layers composite wall-2 throughout year

Summary of Obtained Results

The numerical data obtained from ANSYS confirms the following:

1. **Heat Flux Reduction:** The hybrid wall assembly achieved a significantly lower heat flux compared to the conventional wall, validating the design's energy-saving potential.
2. **Material Synergy:** The 60:40 ratio Water Hyacinth-Cement mix proved to be an effective low-cost insulator which is nearly 10 times more resistant than standard cement plastering.
3. **Optimization Potential:** The results indicate that even thin layers (10mm) of bio-insulation and glass wool can change the thermal profile of standard concrete buildings.

After running the full-scale simulations in **OpenStudio** using the **Thiruvananthapuram** weather data, we compared the annual energy performance of several different wall configurations. This allowed us to see how adding more layers of insulation impacted the total cooling load of our 50 m² office building.

Wall Configuration	Description	Total Site Energy (Annual)	Total Site EUI
Conventional Wall	Plaster + Concrete + Plaster	14,592 kWh	291.83 kWh/m ²
4-Layer Composite	Hyacinth + Glass-Wool + Concrete + Plaster	10,525 kWh	210.50 kWh/m ²
5-Layer Wall-1	2x Hyacinth + Glass-Wool + Concrete + Plaster	9,986 kWh	199.72 kWh/m ²
5-Layer Wall-2	Hyacinth + 2x Glass-Wool + Concrete + Plaster	9,714 kWh	194.28 kWh/m ²

Table. 1.5 Comparison of Wall Configuration Performance

DISCUSSION

In this study, we analyzed the thermal performance of several building wall configurations to address the high energy demands in the tropical climate of Thiruvananthapuram. Using **ANSYS**, we first conducted steady-state thermal simulations to observe how different material layers resist heat flux at a fundamental level. Our findings from the **OpenStudio** and **EnergyPlus** annual energy simulations provided a clear numerical comparison between conventional construction and our engineered composite designs. The baseline conventional wall, consisting of standard concrete and plaster, resulted in an annual energy consumption of **14,592 kWh** with a Site EUI of **291.83 kWh/m²**. In comparison, the 5-layer variations achieved the highest theoretical efficiency; the configuration with two layers of water hyacinth (Type A) reduced consumption to **9,986 kWh**, while the configuration with two layers of glass-wool (Type B) reached the lowest energy demand at **9,714 kWh**, representing a 33.4% saving.

Despite the superior energy performance of the 5-layer designs, we have selected the **4-layer composite wall** (Cement-Water Hyacinth, Glass-Wool, Concrete, and Plaster) as our final recommended configuration. From a mechanical engineering perspective, this choice was made by carefully balancing thermal efficiency with **economic and practical aspects**. The 4-layer wall successfully lowered the annual energy consumption to **10,525 kWh** (Site EUI of **210.50 kWh/m²**), achieving a substantial **27.8% reduction** in energy usage compared to the conventional wall. While the 5-layer walls offer incremental gains, they also involve higher material costs, increased labor complexity, and greater total wall thickness. By selecting the 4-layer system, we provide a solution that maximizes energy conservation while remaining cost-effective and feasible for real-world construction. This project demonstrates that integrating sustainable bio-composites like water hyacinth with industrial insulators can significantly reduce the carbon footprint and operational costs of modern buildings.

CONCLUSION

In conclusion, this project has successfully demonstrated the importance of multi-layer composite walls in enhancing the thermal efficiency of buildings. Through a combination of fundamental steady-state thermal analysis in **ANSYS** and whole-building energy modelling in **OpenStudio**, we validated that using insulation layers specifically **Glass-Wool** and our sustainable **Cement-Water Hyacinth composite** effectively we can reduce the heat flux. Our simulations revealed that while the 5-layer configurations yielded the highest theoretical energy savings—with the 2-layer Water Hyacinth design (Type A) reaching **9,986 kWh** and the 2-layer Glass-Wool design (Type B) reaching **9,714 kWh**—the **4-layer composite wall** emerged as the most balanced engineering solution. By reducing annual site energy consumption from **14,592 kWh** to **10,525 kWh**, the 4-layer design achieved a substantial **27.8% energy reduction**. While the 5-layer walls offer slightly better performance (up to 33.4% savings), we have concluded that the 4-layer wall is the superior choice when **considering economic aspects**, as it offers significant savings with lower material costs, reduced wall thickness, and simpler construction. Ultimately, this study proves that by prioritizing sustainable material selection and implementing advanced computational tools, we can design energy-efficient building envelopes that significantly lower operational costs and the carbon footprint of the construction sector in Kerala.

Feature	Conventional Wall	Composite Wall	Impact
Build Cost	Baseline Price	Lower than Conventional	Immediate Profit/Savings
Annual Energy Use	14,592 kWh	10,525 kWh	4,067 kWh saved per year
Unit Rate (>500 units)	₹8.70 / unit	₹8.70 / unit	Current 2026 KSEB Rate
Annual Electricity Cost	₹1,26,950	₹91,567	₹35,383 saved every year
Thermal Flux	73.16 W/m ²	17.59 W/m ²	76% better heat rejection

Table. 1.6 Economic and Thermal Performance Comparison

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